6-8 Digitization of Images

Point!

1 Digitization of Images

(1) ([1]**Pixel**) : The smallest unit that composes an image.
A digital image is represented by an arrangement of pixels.

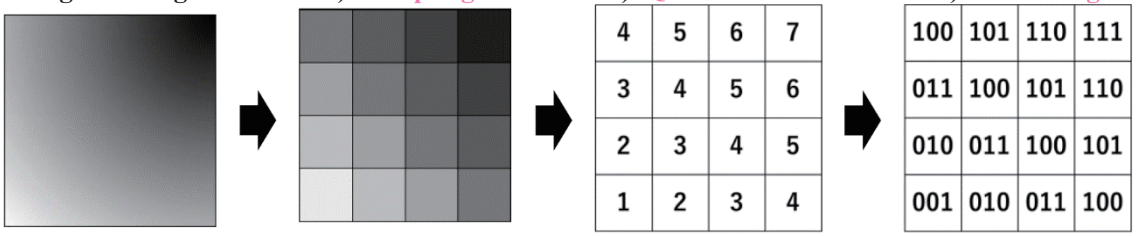
(2) Procedure for Digitization of Images
Images captured by digital cameras or image scanners are digitized and imported into a computer through the following process.

1) ([2]**Sampling**) : The image is divided into pixels, and the representative brightness is extracted.
- ([3]**Resolution**) : The degree of fineness of pixels when sampling.
The unit of resolution is ([4]**dpi**).
It is sometimes expressed as vertical pixels × horizontal pixels.

2) ([5]**Quantization**) : Converts the brightness of each pixel into numerical values divided into several levels.
- ([6]**Gradation**) : The level value that represents the intensity of color per pixel. The gradation is determined by the number of bits allocated to each pixel.
<Example> At 256 gradients (8 bits), it represents 256 levels from 0 to 255.

3) ([7] **Encoding**) : Expressing quantized values in binary numbers (0 and 1).

Original Image → **1) Sampling** → **2) Quantization** → **3) Encording**



2) Quantization			
4	5	6	7
3	4	5	6
2	3	4	5
1	2	3	4

3) Encording			
100	101	110	111
011	100	101	110
010	011	100	101
001	010	011	100

(3) Digitization and Data Volume
[1] Data amount of an image [bits]
= ([8] **Number of pixels [vertical pixels × horizontal pixels) × Number of bits for color information**)
[2] The ([9]**higher**) the values of resolution and gradation, the smoother the image obtained and the image quality ([10]**improves**), but the amount of data ([11]**increases**).

(4) Raster Format and Vector Format
[1] ([12]**Raster format**) : An arrangement of pixels used to represent text or shapes. It appears jagged when enlarged (this appearance is called ([13]**jaggies**)).
Images are drawn using ([14]**painting software**).
[2] ([15]**Vector format**) : A representation that includes information about coordinates of points composing the image, as well as the angles and thickness of the lines connecting them. Images are drawn using ([16]**drawing software**).

2 Color Representation

(1) ([17]**Three primary colors of light**) : The three colors of red, green, and blue. When these colors of light are mixed, the brightness increases and approaches ([18]**white**).

(2) ([19]**Three primary colors of pigment**) : The three colors of cyan, magenta, and yellow. Mixing these colors will darken them and bring them closer to ([20]**black**).

(3) ([21]**Additive color mixing**) : A method of representing colors by combining the three primary colors of light. Used in televisions and computer displays.

(4) ([22]**Subtractive color mixing**) : A method of representing colors by combining the three primary colors of pigment. Used in color printers.

(5) ([23]**24-bit full color**) : Each color of red, green, and blue is represented in 256 levels (2^8 = 8 bits) × 3 = 24 bits.

Warm Up

Answer the following questions.

(1) Rearrange the following steps **A** to **C** in the order of converting an image into digital data.
- **A** Replace the brightness levels of each region with integer values according to a fixed rule.
- **B** Divide the image into evenly spaced grid sections.
- **C** Represent integers in binary.

(2) Choose all statements that correctly describe the digital representation of images from the options **A** to **D**, and answer using the letters.
- **A** The raster format appears as having noticeably jagged edges when the image is enlarged.
- **B** The raster format is suitable for expressing images with clear contours, such as logos.
- **C** Vector format represents images by incorporating information such as the coordinates of points and the thickness of lines.
- **D** Painting software is used to create vector graphics.

(3) What is the data amount of a 24-bit full-color image with a resolution of 1,280 × 720 pixels in megabytes? For this question, assume that 1 KB = 1,000 B and 1 MB = 1,000 KB, and round your answer to two decimal places.

Explanation

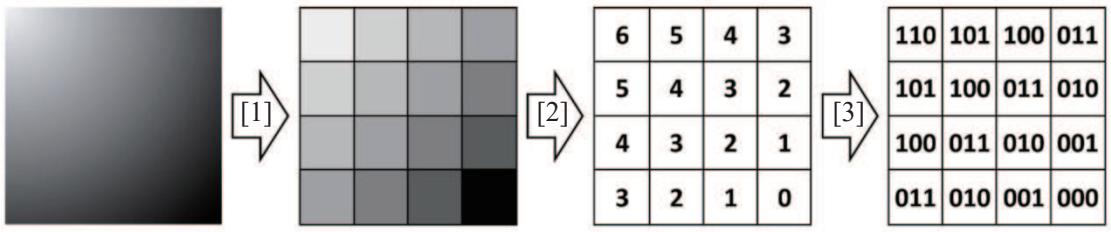
(1) **B → A → C**

(2) **A**, **C**

(3) The amount of data in the image is,
1,280 × 720 × 24 [bits] = 22,118,400 [bits]
Since 1 B equals 8 bits, 22,118,400 ÷ 8 = 2,764,800 [B].
1 KB = 1,000 B, 1 MB = 1,000 KB, therefore,
2,764,800 [B] ÷ 1,000 ÷ 1,000 = 2.7648 [MB] ≈ 2.76 [MB] 2.76 MB

Try

1 The following diagram illustrates the process of converting a monochrome image into digital data. Choose the appropriate operations for steps **[1]** to **[3]** from [Group A], and then choose the name of each operation from [Group B]. Answer using the numbers.



[Group A] **A** Replace the brightness levels of each region with integer values according to a fixed rule.
B Divide the image into evenly spaced grid sections.
C Represent integers in binary.

[Group B] **A** Quantization **B** Encoding **C** Sampling

2 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] and [2] with the appropriate term and number.

> To represent colors, a computer display uses a combination of dots in red, green, and blue. These are known as the ([1]). The intensity of each of the primary colors is expressed in gradations. When representing one color with 8 bits, there are ([2]) gradients.

(2) Complete the following sentences by filling in the blanks [1] and [2] with the appropriate terms.

> Images processed by computers can be divided into two formats: the ([1]) format, which treats the image as a collection of tiny dots, and the ([2]) format, which uses a combination of mathematical expressions to represent the coordinates of points, lines and surfaces connecting them, and color information.

(3) Choose one statement that is incorrect as an explanation of the digital representation of images from the options **A** to **D**.

A The unit dpi represents the degree of pixel density.
B The number of gradations depends on the number of bits allocated per pixel.
C The smaller the resolution and gradation values, the smoother the image obtained.
D Gradation indicates the number of steps or levels of color or brightness.

3 Answer the following questions. In these questions, assume that 1 KB = 1,000 B, and 1 MB = 1,000 KB.

(1) What is the data amount of an image in kilobytes with a resolution of 800 × 600 pixels, where each pixel has 8 bits?

(2) What is the data amount of a 24-bit full-color image with a resolution of 720 × 480 pixels in megabytes? Round your answer to two decimal places.

(3) What is the data amount of a 24-bit full-color image with a resolution of 3,820 × 2,160 pixels in megabytes? Round your answer to two decimal places.

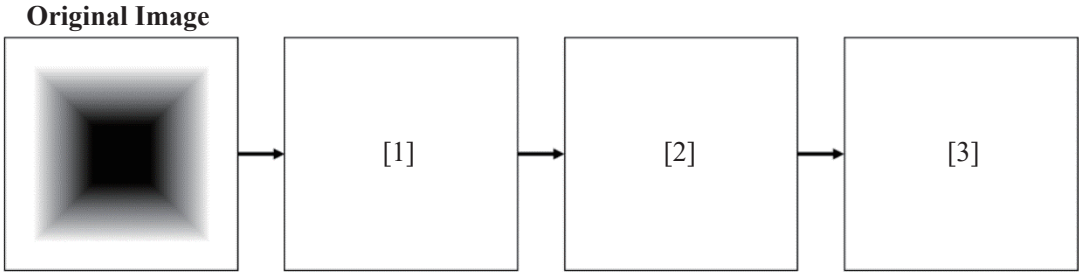
Exercise

1 Answer the following questions.

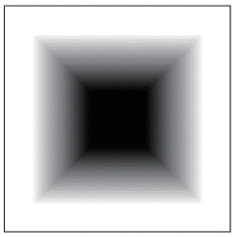
(1) Choose the correct flow for digitizing an image captured by a digital camera from the options **A** to **D**, and answer using the letter.

A Quantization → Encoding → Sampling
B Quantization → Sampling → Encoding
C Encoding → Quantization → Sampling
D Sampling → Quantization → Encoding

(2) The following diagram illustrates the process of converting the original image into digital data. Choose the diagram that best fits into the blanks [1] to [3] from the options **A** to **D**, and answer using the letters.



A



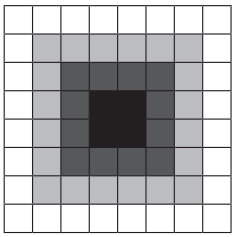
B

0	0	0	0	0	0	0	0
0	1	1	1	1	1	1	0
0	1	2	2	2	2	1	0
0	1	2	3	3	2	1	0
0	1	2	3	3	2	1	0
0	1	2	2	2	2	1	0
0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0

C

00	00	00	00	00	00	00	00
00	01	01	01	01	01	01	00
00	01	10	10	10	10	01	00
00	01	10	11	11	10	01	00
00	01	10	11	11	10	01	00
00	01	10	10	10	10	01	00
00	01	01	01	01	01	01	00
00	00	00	00	00	00	00	00

D



2 Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **D** below.

> In color printers, a variety of colors are expressed by combining the three colors of cyan, magenta, and yellow, which are the ([1]). When these colors are mixed together, they approach the color ([2]). On the other hand, computer displays represent all colors by combining red, green, and blue, which are the ([3]).

A Three primary colors of light
B Three primary colors of pigment
C White
D Black

3 Answer the following questions. In these questions, assume that 1 KB = 1,000 B, and 1 MB = 1,000 KB.

(1) What is the data amount of an image in kilobytes with a resolution of 1,080 × 720 pixels, where each pixel has 8 bits? Round your answer to the nearest whole number.

(2) What is the data amount of a 24-bit full-color image with a resolution of 320 × 480 pixels in kilobytes? Round your answer to the nearest whole number.

(3) What is the data amount of a 24-bit full-color image with a resolution of 1,920 × 1,080 pixels in megabytes? Round your answer to two decimal places.